ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Sound Wave

English Medium

EXERCISE (O-1)

A. Displacement and pressure wave equation

1. A tuning fork makes 256 vibrations per second in air. When the velocity of sound is 330 m/s, then wave length of tone emitted is

(A) 0.56 m (B) 0.89 m (C) 1.11 m (D) 1.29 m

SW0152

2. A sound wave of wavelength 90 cm in glass is refracted into air. If the velocity of sound in glass is 5400 m/sec, the wavelength of the wave in air is (The velocity of sound in air is 330 m/s)

(A) 55 cm (B) 5.5 cm (C) 55 m (D) 5.5 m

SW0153

3. A sound wave has a wavelength of 3.0 m. The distance from a compression center to the adjacent rarefaction center is :-

(A) 0.75 m (B) 1.5 m

(C) 3.0 m (D) need to know wave speed

SW0030

4. You are listening to an "A" note played on a violin string. Let the subscript "s" refer to the violin string and "a" refer to the air. Then :-

(A) $f_s = f_a$ but $\lambda_s \neq \lambda_a$ (B) $f_s = f_a$ and $\lambda_s = \lambda_a$

(C) $\lambda_s = \lambda_a$ but $f_s \neq f_a$ (D) $\lambda_s \neq \lambda_a$ and $f_s \neq f_a$

SW0031

5. The fig.(i) shows the graphical representation of the air molecules in a tube of air (length = L) at atmospheric pressure on the absolute pressure P(x) graph. Which one of the following pictures corresponds to the absolute pressure P(x) graph of fig. (ii).

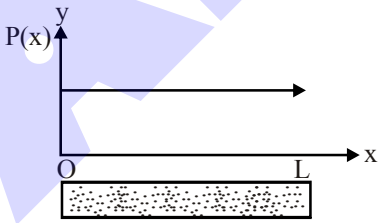


Figure (i)

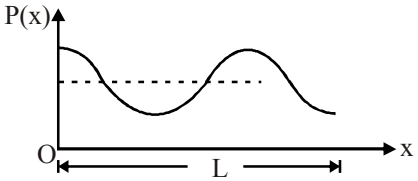
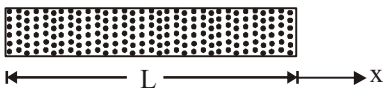
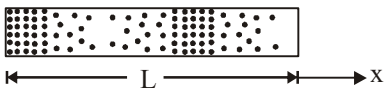
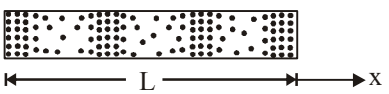
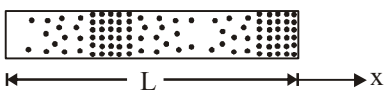


Figure (ii)

(A) 

(B) 

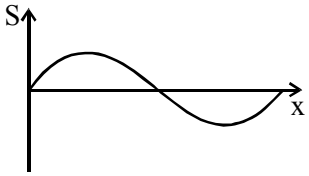
(C) 

(D) 

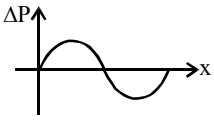
SW0032

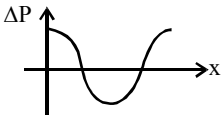
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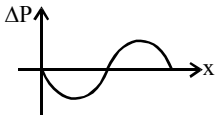
6. If a sound wave is travelling and snap shot at t = 0 is as shown in figure.

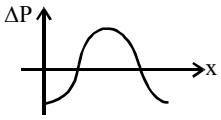


Choose snapshot of pressure variation.

(i) 

(ii) 

(iii) 

(iv) 

(A) For wave travelling towards right or left (i) is correct.

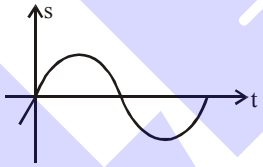
(B) For wave travelling towards right graph (iv) and for wave travelling towards left graph (iv) is correct.

(C) For wave travelling towards right graph (i) and for wave travelling towards left graph (iii) is correct

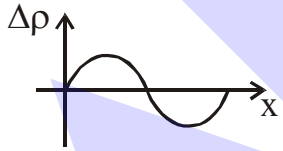
(D) For wave travelling towards right or left (ii) is correct.

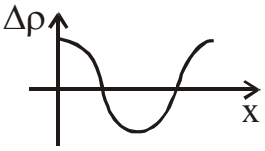
SW0033

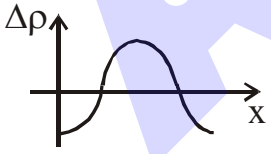
7. A sound waves is travelling towards right and its s-t graph is as shown for x = 0.

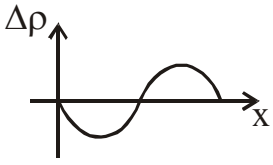


What will be the variation in density vs x graph at t = T/4 :-

(A) 

(B) 

(C) 

(D) 

SW0034

B. Properties of sound

8. Two sound waves move in the same direction in the same medium. The pressure amplitudes of the waves are equal but the wavelength of the first wave is double the second. Let the average power transmitted across a cross-section by the first wave be P_1 and that by the second wave be P_2. Then

(A) $P_1 = P_2$ (B) $P_1 = 4P_2$ (C) $P_2 = 2P_1$ (D) $P_2 = 4P_1$

SW0154

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9. A sound level I is greater by 3.0103 dB from another sound of intensity 10 nW cm^{-2}. The absolute value of intensity of sound level I in Wm^{-2} is (nearly) :

(A) 2.0×10^{-7} (B) 2×10^{-4} (C) 2.0×10^{-2} (D) 2.5×10^{-2}

SW0155

10. A person is talking in a small room and the sound intensity level is 60 dB everywhere within the room. If there are eight people talking simultaneously in the room, what is the sound intensity level ?

(A) 60 dB (B) 69 dB (C) 74 dB (D) 81 dB

SW0156

11. A point source emits sound equally in all directions in a non-absorbing medium. Two points P and Q are at a distance of 9m and 25m respectively from the source. The ratio of the amplitude of waves at P and Q is

(A) 3/5 (B) 5/3 (C) 9/25 (D) 25/9

SW0157

12. A blast gives a sound of intensity $0.8\ W/m^2$ and frequency 1 kHz . If the density of air is 1.3 kg/m^3 and speed of sound in air is 330 m/s, the amplitude of the sound wave is approximately

(A) $5\times10^{-6}m$ (B) $9.7\times10^{-6}m$ (C) $15\times10^{-6}m$ (D) $20\times10^{-6}m$

SW0158

13. Velocity of sound in a diatomic gas is 300 m/s. What is its rms velocity ?

(A) 400 m/s (B) 40 m/s (C) 430 m/s (D) 300 m/s

SW0159

14. The speed of sound in a mixture of 1 mole of helium and 2 moles of oxygen at 27°C is

(A) 400 m/s (B) 800 m/s (C) 1200 m/s (D) 600 m/s

SW0160

15. The speed of sound waves in dry air at NTP is 332 m/sec. Assuming air to be composed of oxygen and nitrogen in the ratio 1 : 4 and the densities of oxygen and nitrogen at NTP to be in the ratio 8 : 7, the velocity of sound in oxygen will be equal to

(A) 314.7 m/sec (B) 350 m/sec (C) 300 m/sec (D) 360 m/sec

SW0161

16. It is found that an increase in pressure of 100 kPa causes a certain volume of water to decrease by 5×10^{-3} percent of its original volume. Then the speed of sound in the water is about (density of water 10^3 kg/m^3)

(A) 330 m/s (B) 1414 m/s (C) 1732 m/s (D) 2500 m/s

SW0162

17. A point source of sound is located somewhere along the x-axis. Experiments show that the same wave front simultaneously reaches listeners at x = –8 m and x = +2.0 m.

A third listener is positioned along the positive y-axis. What is her y-coordinate (in m) if the same wave front reaches her at the same instant as it does the first two listeners ?

(A) 4 (B) 3 (C) 2 (D) 5

SW0035

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18. Two monatomic ideal gases 1 and 2 of molecular masses m_1 and m_2 respectively are enclosed in separate container kept at the same temperature. The ratio of the speed of sound in gas 1 to that in gas 2 is given by

[JEE 2000 (Scr)]

(A) $\sqrt{\frac{m_1}{m_2}}$ (B) $\sqrt{\frac{m_2}{m_1}}$ (C) $\frac{m_1}{m_2}$ (D) $\frac{m_2}{m_1}$

SW0036

19. A firecracker exploding on the surface of a lake is heard as two sounds a time interval *t* apart by a man on a boat close to water surface. Sound travels with a speed *u* in water and a speed *v* in air. The distance from the exploding firecracker to the boat is

(A) $\frac{uvt}{u+v}$ (B) $\frac{t(u+v)}{uv}$ (C) $\frac{t(u-v)}{uv}$ (D) $\frac{uvt}{u-v}$

SW0037

20. The speed of longitudinal wave is 100 times the speed of transverse wave in a taut brass wire. If the Young's modulus of brass is 1.0×10^{11} N/m^2, the stress in wire is :-

(A) 1.0×10^7 N/m^2 (B) 1.0×10^6 N/m^2

(C) 1.0×10^5 N/m^2 (D) 1.0×10^8 N/m^2

SW0038

21. A is singing a note and at the same time B is also singing a note with 1/8th the frequency of A. The energies of the two sounds are equal. The displacement amplitude of the note of B is:

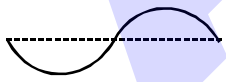
(A) same as that of A (B) twice that of A

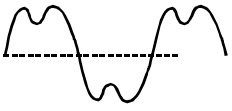
(C) four times that of A (D) eight times that of A

SW0040

22. A microphone is connected to an oscilloscope. The diagram shows the trace on the screen when the microphone receives a pure note. Which trace can be obtained when a musical instrument produces a note of the same pitch but of a different quality ?



(A) 

(B) 

(C) 

(D) 

SW0041

23. A sound absorber attenuates the sound level by 20 dB. The intensity decreases by a factor of-

[AIEEE - 2007]

(A) 1000 (B) 10000 (C) 10 (D) 100

SW0046

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C. Interference of waves and beats

24. Four simple harmonic vibrations

$y_1 = 8\cos\omega t,\quad y_2 = 4\cos\left(\omega t + \frac{\pi}{2}\right),\quad y_3 = 2\cos(\omega t + \pi),\quad y_4 = \cos\left(\omega t + \frac{3\pi}{2}\right)$ are superposed on each other. The resulting amplitude and phase are respectively

(A) $\sqrt{45}$ and $\tan^{-1}(1/2)$ (B) $\sqrt{45}$ and $\tan^{-1}(1/3)$

(C) $\sqrt{75}$ and $\tan^{-1}(1/2)$ (D) $\sqrt{75}$ and $\tan^{-1}(1/3)$ **SW0163**

25. The ratio of maximum to minimum intensity due to superposition of two waves is $\frac{49}{9}$. Then the ratio of the intensity of component waves is :-

(A) $\frac{25}{4}$ (B) $\frac{16}{25}$ (C) $\frac{4}{49}$ (D) $\frac{9}{49}$

SW0051

26. Two waves of sound having intensities I and 4I interfere to produce interference pattern. The phase difference between the waves is $\pi/2$ at point A and π at point B. Then the difference between the resultant intensities at A and B is

(A) 2I (B) 4I (C) 5I (D) 7I

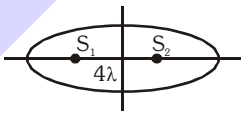
SW0052

27. Three coherent waves of equal frequencies having amplitude 10 μm, 4μm and 7 μm respectively, arrive at a given point with successive phase difference of $\pi/2$. The amplitude of the resulting wave in μm is given by

(A) 5 (B) 6 (C) 3 (D) 4

SW0053

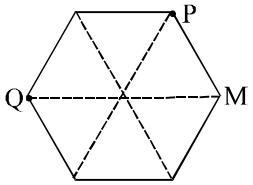
28. S_1, S_2 are two coherent sources of sound located along x – axis separated by 4λ where λ is wavelength of sound emitted by them. Number of maxima located on the elliptical boundary around it will be :



(A) 16 (B) 12 (C) 8 (D) 4

SW0056

29. Two sound source emitting sound of wavelength 1 m are located at points P and Q as shown in figure. All sides of the polygon are equal and of length 1m. The intensity of sound at M due to both the individual sources is I_0. What will be the intensity of sound at point M when both the sources are on.

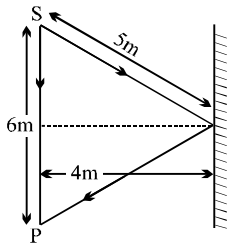


(A) $4I_0$ (B) $2I_0$ (C) I_0 (D) $(1/2)I_0$

SW0057

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30. A person standing at a distance of 6 m from a source of sound receives sound wave in two ways, one directly from the source and other after reflection from a rigid boundary as shown in the figure. The maximum wavelength for which, the person will receive maximum sound intensity, is



(A) 4 m (B) $\frac{16}{3}$ m (C) 2 m (D) $\frac{8}{3}$ m

SW0058

31. In Quincke's tube a detector detects minimum intensity. Now one of the tube is displaced by 5 cm. During displacement detector detects maximum intensity 10 times, then finally a minimum intensity (when displacement is complete). The wavelength of sound is:

(A) 10/9 cm (B) 1 cm (C) 1/2 cm (D) 5/9 cm

SW0055

32. Beats are heard when the A strings of two violins are played. The beat frequency decreases as the tension in the A string of violin 1 is slowly increased. Which of the following statement is correct?

(A) the fundamental frequency of the A string in violin 1 is less than that for violin 2

(B) the fundamental frequency of the A string in violin 1 is greater than that for violin 2

(C) the fundamental frequency of the A string in violin 1 may be greater or less than that for violin 2 depending on the linear mass densities of the two strings.

(D) None of these

SW0059

33. Beats occur because of

(A) interference (B) reflection

(C) refraction (D) Doppler effect

SW0164

34. When beats are produced by two waves, viz., $y_1 = A\sin 1000\pi t$ and $y_2 = A\sin 1008\pi t$, the beat frequency will be

(A) 4 Hz (B) 8 Hz (C) $4\pi\, Hz$ (D) $8\pi\, Hz$

SW0165

35. A tuning fork of frequency 512 Hz is vibrated with a sonometer wire and 6 beats per second are heard. The beat frequency reduces if the tension in the string is slightly increased. The original frequency of vibration of the string is

(A) 506 Hz (B) 512 Hz (C) 518 Hz (D) 524 Hz

SW0166

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36. When beats are produced by two progressive waves of nearly the same frequency, which one of the following is correct?

(A) The particles vibrate simple harmonically, with the frequency equal to the difference in the component frequencies.

(B) The amplitude of vibration at any point changes simple harmonically with a frequency equal to the difference in the frequencies of the two waves.

(C) The frequency of beats depends upon the position, where the observer is

(D) The frequency of beats changes as the time progresses

SW0167

37. A closed organ pipe and an open pipe of same length produce 4 beats when they are set into vibrations simultaneously. If the length of each of them were twice their initial lengths, the number of beats produced will be [Assume same mode of vibration in both cases]

(A) 2 (B) 4 (C) 1 (D) 8

SW0168

38. The speed of sound in a gas in which two waves of wavelengths 50 cm and 50.4 cm produce 6 beats per second is

(A) 300 m/s (B) 340 m/s (C) 375 m/s (D) 378 m/s

SW0169

39. 56 tuning forks are so arranged in series that each fork gives 4 beats per sec with the previous one. The frequency on the last fork is 3 times that of the first. The frequency of the first fork is

(A) 110 (B) 56 (C) 60 (D) 52

SW0170

40. When two tuning forks (fork 1 and fork 2) are sounded simultaneously, 4 beats per second are heard. Now, some tape is attached on the prong of the fork 2. When the tuning forks are sounded again, 6beats per second are heard. If the frequency of fork 1 is 200 Hz, then what was the original frequency of fork 2 ? **[AIEEE - 2005]**

(A) 200 Hz (B) 202 Hz (C) 196 Hz (D) 204 Hz

SW0171

D. Standing waves

41. When a tuning fork vibrates, the waves produced in the fork are

(A) longitudinal (B) transverse

(C) progressive (D) stationary

SW0172

42. The three lowest frequencies (in Hz) with which a 20 cm long pipe, closed at one end, can vibrate are (speed of sound = 340 m/s)

(A) 425,850,1275 (B) 425,1275,2125

(C) 900,1800,2700 (D) 900,2700,4500

SW0173

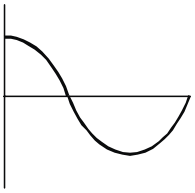
43. Velocity of sound in air is 320 m/s. A pipe closed at one end has a length of 1 m. Neglecting end corrections, the air column in the pipe can not resonate for sound of frequency

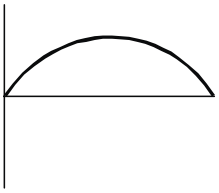
(A) 80 Hz (B) 240 Hz (C) 320 Hz (D) 400 Hz

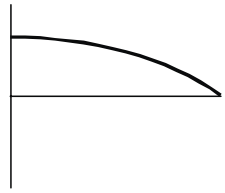
SW0174

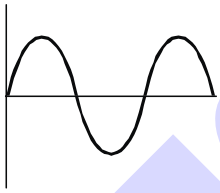
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44. Which of the figures, shows the pressure difference from regular atmospheric pressure for an organ pipe of length L closed at one end, corresponds to the 1st overtone for the pipe?

(A) 

(B) 

(C) 

(D) 

SW0047

45. In an organ pipe whose one end is at x = 0, the pressure is expressed by

$p = p_0 \cos \frac{3\pi x}{2} \sin 300\pi t$ where x is in meter and t in sec. The organ pipe can be

(A) closed at one end, open at another with length = 0.5m

(B) open at both ends, length = 1m

(C) closed at both ends, length = 2m

(D) closed at one end, open at another with length = $\frac{2}{3}$ m

SW0048

46. In a long cylindrical tube, the water level is adjusted and the air column above it is made to vibrate in unison with a vibrating tuning fork kept at the open end. Maximum sound is heard when the air column lengths are equal to

(A) $\frac{\lambda}{4}, \frac{\lambda}{2}, \frac{3\lambda}{4}$ (B) $\frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}$ (C) $\frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}$ (D) $\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$

SW0175

47. If l_1 and l_2 are the lengths of air column for the first and second resonance when a tuning fork of frequency n is sounded on a resonance tube, then the distance of the displacement antinode from the top end of the resonance tube is:

(A) $2(l_2 - l_1)$ (B) $\frac{1}{2}(2l_1 - l_2)$ (C) $\frac{l_2 - 3l_1}{2}$ (D) $\frac{l_2 - l_1}{2}$

SW0049

48. A tuning fork of frequency 340 Hz is vibrated just above a cylindrical tube of length 120 cm. Water is slowly poured in the tube. If the speed of sound in air is 340 m/s, then the minimum height of water required for resonance is

(A) 25 cm (B) 45 cm (C) 75 cm (D) 95 cm

SW0176

node06\B09C-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_Th+Ex_p65

49. A student is experimenting with resonance tube apparatus in Physics lab to find the speed of sound at room temperature. He got resonating lengths of air column as 17 cm and 51 cm, using tuning fork of frequency 512 Hz. Find speed of sound at room temperature and specify, whether the side water reservoir was moved upward or downward to obtain the second resonance (51 cm)?

(A) 348 m/s, downwards

(B) 348 m/s, upwards

(C) 332 m/s, downwards

(D) 332 m/s, upwards

SW0050

50. In the experiment to determine the speed of sound using a resonance column, **[JEE 2007]**

(A) prongs of the tuning fork are kept in a vertical plane

(B) prongs of the tuning fork are kept in a horizontal plane

(C) in one of the two resonances observed, the length of the resonating air column is close to the wavelength of sound in air

(D) in one of the two resonances observed, the length of the resonating air column is close to half of the wavelength of sound in air

SW0177

51. While measuring the speed of sound by performing a resonance column experiment, a student gets the first resonance condition at column length of 18 cm during winter. Repeation the same experiment during sumer, student measures the column length to be x cm for the second resonance. Then

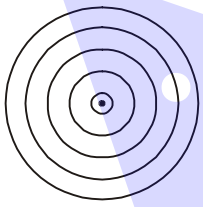
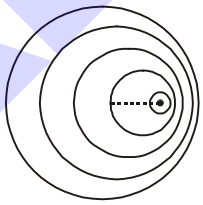
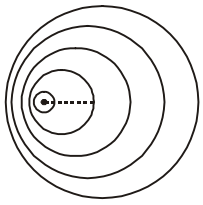
[AIEEE - 2008]

(A) $18 > x$ (B) $x > 54$

(C) $54 > x > 36$ (D) $36 > x > 18$ **SW0178**

E. Doppler effect

52. If the source is moving towards right, wave front of sound waves get modified to -

(A)  (B)  (C)  (D) None of these

SW0179

53. A source when at rest in a medium produces waves with a velocity v and a wavelength of λ. If the source is set in motion with a velocity v_s what would be the wavelengths produced directly in front of the source?

(A) $\lambda\left(1-\frac{v_s}{v}\right)$ (B) $\lambda\left(1+\frac{v_s}{v}\right)$ (C) $\lambda\left(1+\frac{v}{v_s}\right)$ (D) $\frac{\lambda v}{v+v_s}$

SW0061

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_TH-Ex.p65

54. A source of sound S having frequency f. Wind is blowing from source to observer O with velocity u. If speed of sound with respect to air is C, the wavelength of sound detected by O is:

(A) $\frac{C+u}{f}$ (B) $\frac{C-u}{f}$ (C) $\frac{C(C+u)}{(C-u)f}$ (D) $\frac{C}{f}$

SW0062

55. A train moving towards a hill at a speed of 72 km/hr sounds a whistle of frequency 500 Hz. A wind is blowing from the hill at a speed of 36 km/ hr. If the speed of sound in air is 340 m / s, the frequency heard by a man on the hill is

(A) 532.25 Hz. (B) 565.0 Hz.

(C) 516.1 Hz. (D) none of the above.

SW0063

56. A siren placed at a railway platform is emitting sound of frequency 5 kHz. A passenger sitting in a moving train A records a frequency of 5.5 kHz while the train approaches the siren. During his return journey in a different train B he records a frequency of 6.0 kHz while approaching the same siren. The ratio of the velocity of train B to that of train A is :-

(A) 242/252 (B) 2 (C) 5/6 (D) 11/6

SW0066

57. A police car with a siren of frequency 8 kHz is moving with uniform velocity 36 km/hr towards a tall building which reflects the sound waves. The speed of sound in air is 320 m/s. The frequency of the siren heard by the car driver is **[JEE-2011]**

(A) 8.50 kHz (B) 8.25 kHz (C) 7.75 kHz (D) 7.50 kHz

SW0089

58. A student sees a jet plane flying from east to west. When the jet is seen just above his head, the sound of jet appears to reach him making angle 60° with the horizontal from east. If the velocity of sound is v, then that of the jet plane is

(A) $2v$ (B) $(\sqrt{3}/2)v$ (C) $(2/\sqrt{3})v$ (D) $v/2$

SW0180

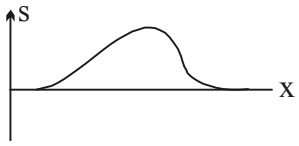
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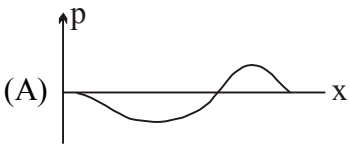
EXERCISE (O-2)

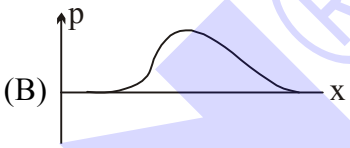
SINGLE CORRECT TYPE QUESTIONS

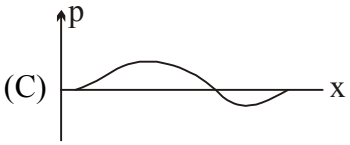
A. Displacement and pressure wave equation

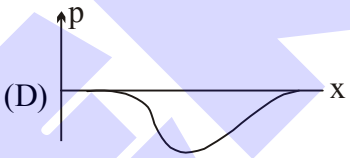
1. For displacement(s)-x graph shown for a sound wave, select appropriate excess pressure(p)-x graph.



(A) 

(B) 

(C) 

(D) 

SW0181

2. Mark the **INCORRECT** statement about longitudinal pressure wave.

(A) The density variation and the longitudinal velocity of the particles have their maxima and minima at the position of equilibrium of the particles.

(B) At the position of maximum displacement the density is at the ambient (environment) value.

(C) Greatest compression occurs at the same time and place where the velocity of the particles is maximum and in direction of wave propagation.

(D) Greatest rarefaction occurs at the same time and place where the velocity of the particles is maximum and in direction of wave propagation. **SW0182**

B. Properties of sound

3. How long will it take by sound waves to travel a distance l between points A and B if the air temperature between them varies linearly from T_1 to T_2? The velocity of sound in air is given by $v = \alpha\sqrt{T}$ where α is a constant.

(A) $\dfrac{2l}{\alpha\sqrt{T_1 T_2}}$ (B) $\alpha l\sqrt{\dfrac{T_1}{T_2}}$ (C) $\left(\sqrt{T_1 + T_2}\right)\alpha l$ (D) $\dfrac{2l}{\alpha\left(\sqrt{T_1} + \sqrt{T_2}\right)}$

SW0183

4. Choose **correct** statement?

(A) Two different acoustic musical instruments can not have same loudness

(B) Two different acoustic musical instruments can not have same pitch

(C) Two different acoustic musical instruments can not have same quality

(D) Two different acoustic musical instruments can have more than two characteristics same

SW0042

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_TH-Ex.p65

5. A plane transverse wave is propagating in a direction making an angle of 30° with positive x-axis in the x-y plane. Find phase difference between points (0, 0, 0) and (1, 1, 1). Wavelength of the wave is 1m :-

(A) 2π rad (B) $\left(\sqrt{3}+1\right)\pi$ rad (C) $\left(\sqrt{2}+1\right)\pi$ rad (D) None

SW0043

6. Which of the following is the equation of a spherical wave :-

(A) $S = S_0 \sin(Kx - \omega t)$ (B) $S = S_0 \cos(Kx - \omega t)$

(C) $S = (S_0/x) \sin(\omega t - Kx)$ (D) $S = (S_0/x^2) \sin(\omega t - Kx)$

SW0044

7. A note is produced when you blow air across the top of a test tube. Two students were asked about the effect of blowing harder.

Student-A : The pitch of sound would increase.

Student-B : The intensity of sound would increase

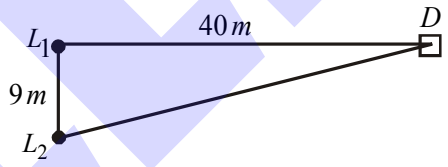
(A) A is correct, B is wrong (B) B is correct, A is wrong

(C) both are correct (D) both are wrong

SW0045

C. Interference of waves and beats

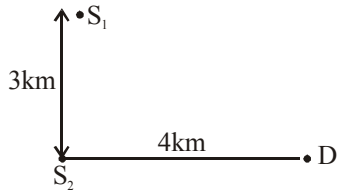
8. Two loudspeakers L_1 and L_2, driven by a common oscillator and amplifier, are arranged as shown. The frequency of the oscillator is gradually increased from zero and the detector at D records a series of maxima and minima. If speed of sound is 330 m/s, then the frequency at which the first maximum is observed is



(A) 165 Hz (B) 330 Hz (C) 495 Hz (D) 660 Hz

SW0184

9. Two point sound source S_1 and S_2 are both have the same power and send out sound waves in the same phase. The wavelength of both the waves is $\frac{48}{5}$ m. The intensity due to S_2 alone at D is 25 W/m^2. The resultant intensity at D is :



(A) 59 W/m^2 (B) 61 W/m^2 (C) 65 W/m^2 (D) None of these

SW0185

node06\B09C-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_Th+Ex_p65

10. The ratio of intensities between two coherent soud sources is 4 : 1. The difference of loudness in dB between maximum and minimum intensities when they interfere in space is:

(A) 10 log 2 (B) 20 log 3 (C) 10 log 3 (D) 20 log 2

SW0054

11. Two bodies producing progressive waves given by $y_1 = 4\sin 400\pi t$ and $y_2 = 3\sin 404\pi t$, where t is in seconds, are situated near the ears of a person. The person will hear

(A) 2 beats per second with intensity ratio 4/3 between maxima and minima

(B) 2 beats per second with intensity ratio 49 between maxima and minima

(C) 4 beats per second with intensity ratio 7 between maxima and minima

(D) 4 beats per second with intensity ratio 4/3 between maxima and minima

SW0186

D. Standing waves

12. The three lowest resonant frequencies of a system are 50 Hz, 150 Hz and 250 Hz. The system could be:

(A) a tube of air closed at both ends.

(B) a tube of air open at one end.

(C) a tube of air open at both ends.

(D) a vibrating string with fixed ends

SW0187

13. In a resonance tube experiment, an 80 cm air column is in resonance with a turning fork in first overtone. Which equation can represent correct pressure variation in the air column (x = 0 is the top point of the tube, neglect end correction, speed of sound = 320 m/sec) :-

(A) A sin $\frac{15\pi}{8}x$ cos 600πt (B) A cos $\frac{15\pi}{8}x$ sin 600 πt

(C) A cos $\frac{15\pi}{8}x$ sin 300 πt (D) A sin $\frac{15\pi}{8}x$ sin 300 πt

SW0188

14. A vibrating string of certain length ℓ under a tension T reasonates with a mode corresponding to the first overtone (third harmonic) of an air column of length 75 cm inside a tube closed at one end. The string also generates 4 beats per second when excited along with a tuning fork of frequency n. Now when the tension of the string is slightly increased the number of beats reduces to 2 per second. Assuming the velocity of sound in air to be 340 m/s, the frequency n of the tuning fork in Hz is.

[IIT-JEE 2008]

(A) 344 (B) 336 (C) 117.3 (D) 109.3

SW0189

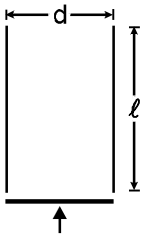
15. An organ pipe P_1, closed at one end and vibrating in its first overtone, and another pipe P_2 open at both ends and vibrating in its third overtone, are in resonance with a given tuning fork. The ratio of the length of P_1 to that of P_2 is

(A) 8/3 (B) 3/8 (C) 1/2 (D) 1/3

SW0190

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_Th+Ex..p65

16. A tube of diameter d and of length ℓ unit is open at both ends. Its fundamental frequency of resonance is found to be ν_1. The velocity of sound in air is 330 m/sec. One end of tube is now closed. The lowest frequency of resonance of tube is ν_2. Taking into consideration the end correction, $\frac{\nu_2}{\nu_1}$ is

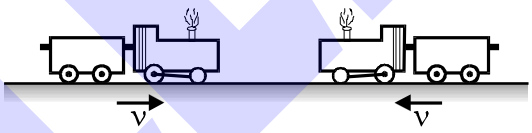


(A) $\frac{(\ell+0.6d)}{(\ell+0.3d)}$ (B) $\frac{1}{2}\frac{(\ell+0.3d)}{(\ell+0.6d)}$ (C) $\frac{1}{2}\frac{(\ell+0.6d)}{(\ell+0.3d)}$ (D) $\frac{1}{2}\frac{(d+0.3\ell)}{(d+0.6\ell)}$

SW0191

E. Doppler effect

17. Two trains move towards each other with the same speed. Speed of sound is 340 ms^{-1}. If the pitch of the tone of the whistle of one when heard on the other changes by 9/8 times, then the speed of each train is:



(A) 2 ms^{-1} (B) 40 ms^{-1} (C) 20 ms^{-1} (D) 100 ms^{-1}

SW0192

18. Two tuning forks A and B lying on opposite sides of observer 'O' and of natural frequency 85 Hz move with velocity 10 m/s relative to stationary observer O. Fork A moves away from the observer while the fork B moves towards him. A wind with a speed 10 m/s is blowing in the direction of motion of fork A. Find the beat frequency measured by the observer in Hz. [Take speed of sound in air as 340 m/s]

(A) 5 (B) 6 (C) 7 (D) 8

SW0193

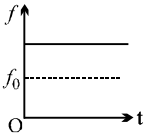
19. A source is moving with constant speed $v_s = 20$ m/sec towards a stationary observer due east of source. Wind is blowing at the speed of 20 m/sec due to 60° north of east. The source is generating of frequency 500 Hz. Then frequency registered by observer is: [Speed of sound in still air = 330 m/sec.]

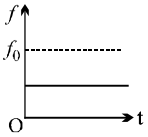
(A) 500 Hz (B) 532 Hz (C) 531 Hz (D) 530 Hz

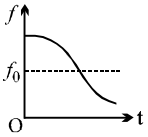
SW0064

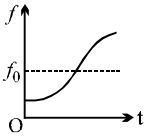
node06\B09C-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_TH+Ex_p65

20. Source and observer both start moving simultaneously from origin, one along x-axis and the other along y-axis with speed of source = twice the speed of observer. The graph between the apparent frequency observed by observer f and time t would approximately be :

(A) 

(B) 

(C) 

(D) 

SW0065

MULTIPLE CORRECT TYPE QUESTIONS

A. Displacement and pressure wave equation

21. The particle displacement of a travelling longitudinal wave is represented by S = S (x, t). The midpoints of a compression zone and an adjacent rarefaction zone are represented by the letter 'C' and 'R'. Which of the following is true?

(A) $|\partial S / \partial x|_C = |\partial S / \partial x|_R$

(B) $|\partial S / \partial t|_C = |\partial S / \partial t|_R = 0$

(C) $(\text{pressure})_C - (\text{pressure})_R = 2\,|\partial S / \partial x|_C$ x Bulk modulus of air.

(D) Particles of air are stationary mid-way between 'C' and 'R'.

SW0069

B. Properties of sound

22. Which of the following statements are wrong about the velocity of sound in air:

(A) decreases with increases in temperature

(B) increases with decrease in temperature

(C) decreases as humidity increases

(D) independent of density of air.

SW0070

23. A small source of sound operates at 1 kHz, 24πW emitting sound uniformly in all directions.The speed of sound is 360m/s in air and density of air is 1.2kg/m^3.

(A) The intensity of sound at a distance 10m from the source is 0.06W/m^2

(B) The intensity of sound at a distance 10m from the source is 0.24W/m^2

(C) The pressure amplitude at distance 10m is 7.2 Pa

(D) The pressure amplitude at distance 10m is 28.8 Pa

SW0194

C. Interference of waves and beats

24. Three coherent source kept along the same line produce intensity I_0 each at point P on this line. When S_1 & S_2 are switched on simultaneously, intensity at point P is $2I_0$. When S_2 and S_3 are switched on simultaneously, intensity at point P is $2I_0$. Then

(A) When S_1 and S_3 are switched on simultaneously, intensity at point P can be $2I_0$

(B) When S_1 and S_3 are switched on simultaneously, intensity at point P can be 0

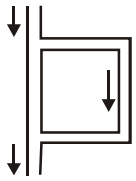
(C) When all 3 sources are switched on simultaneously, intensity at point P can be I_0

(D) When all 3 sources are switched on simultaneously, intensity at point P can be $3I_0$

SW0072

node06\B0BC-BD\Kota\IEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Optics\Eng\02_SW\01_TH+Ex..p65

25. A sound consists of four frequencies → 300 Hz, 900 Hz, 2400 Hz and 4500 Hz. A sound 'filter' is made by passing this sound through a bifurcated pipe as shown. The sound waves have to travel a distance of 50 cm more in the right branch-pipe than in the straight pipe. The speed of sound in air is 300 m/s. Then, which of the following frequencies will be almost completely muffled or "silenced" at the outlet?



(A) 300 Hz (B) 900 Hz (C) 2400 Hz (D) 4500 Hz

SW0073

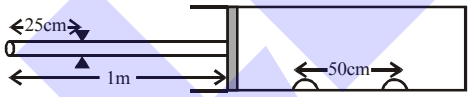
26. Two tuning forks of frequency 250 Hz and 256 Hz produce beats. If a maximum of intensity is observed just now, after how much time the minimum is observed at the same place ?

(A) $\frac{1}{18}$ sec (B) $\frac{1}{4}$ sec. (C) $\frac{1}{3}$ sec. (D) $\frac{1}{12}$ sec.

SW0068

D. Standing waves

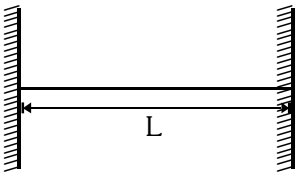
27. In a kundt's tube experiment metallic rod of length 1m is clamped from the distance of 25 cm from one end. If distance between heap of powder 50 cm and speed of sound in air is 330 m/s select correct statement(s) :-

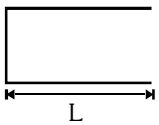


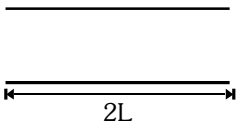
(A) Frequency in which tube is oscillating is 1320 Hz
(B) Frequency in which tube is oscillating is 330 Hz
(C) Velocity of wave in metallic rod is 330 m/s
(D) Velocity of wave in metallic rod is 1320 m/s

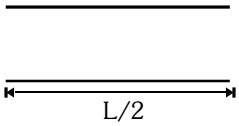
SW0195

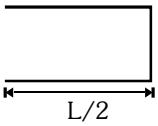
28. Figure shows a streched string of length L and pipes of length L, 2L, L/2 and L/2 in options (A), (B), (C) and (D) respectively. The string's tension is adjusted until the speed of waves on the string equals the speed of sound waves in air. The fundamental mode of oscillation is then set up on the string. In which pipe will the sound produced by the string cause resonance?



(A) 

(B) 

(C) 

(D) 

SW0196

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E. Doppler effect

29. Mark the **CORRECT** statement (s) :-

(A) When two waves of different frequency f_1 and f_2 are superimposed at a point, the variation of intensity takes place with frequency $f_1 - f_2$

(B) When an observer is moving, relative speed of the sound waves with respect to the observer is changed, while due to motion of source, the sound wavelength of the wave changes.

(C) When a high pressure compression pulse reaches water surface in a resonance column, it reflects as a low pressure rarefraction pulse due to a phase change of π

(D) When a tuning fork is struck by a hammer a transverse wave sets in the prong while a longitudinal wave sets in the stem (handle) of tuning fork

SW0197

30. A car moves towards a hill with speed v_c. It blows a horn of frequency f which is heard by an observer following the car with speed v_0. The speed of sound in air is v.

(A) the wavelength of sound reaching the hill is $\frac{v}{f}$

(B) the wavelength of sound reaching the hill is $\frac{v - v_c}{f}$

(C) the beat frequency observed by the observer is $\left(\frac{v + v_o}{v - v_c}\right) f$

(D) the beat frequency observed by the observer is $\frac{2v_c(v + v_o)f}{v^2 - v_c^2}$

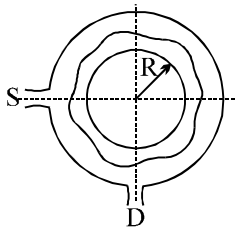
SW0071

COMPREHENSION TYPE QUESTIONS

C. Interference of waves and beats

Paragraph for Question No. 31 to 35

A narrow tube is bent in the form of a circle of radius R, as shown in the figure. Two small holes S and D are made in the tube at the positions right angle to each other. A source placed at S generated a wave of intensity I_0 which is equally divided into two parts : One part travels along the longer path, while the other travels along the shorter path. Both the part waves meet at the point D where a detector is placed



31. If a maxima is formed at the detector then, the magnitude of wavelength λ of the wave produced is given by :-

(A) πR (B) $\frac{\pi R}{2}$ (C) $\frac{\pi R}{4}$ (D) $\frac{2\pi R}{3}$

SW0077

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32. If the minima is formed at the detector then, the magnitude of wavelength λ of the wave produced is given by :-

(A) $2\pi R$ (B) $\frac{3\pi R}{2}$ (C) $\frac{2\pi R}{3}$ (D) $\frac{2\pi R}{5}$

SW0077

33. The maximum intensity produced at D is given by :-

(A) $4I_0$ (B) $2I_0$ (C) I_0 (D) $3I_0$

SW0077

34. The maximum value of λ to produce a maxima at D is given by :-

(A) πR (B) $2\pi R$ (C) $\frac{\pi R}{2}$ (D) $\frac{3\pi R}{2}$

SW0077

35. The maximum value of λ to produce a minima at D is given by :-

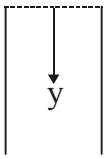
(A) πR (B) $2\pi R$ (C) $\frac{\pi R}{2}$ (D) $\frac{3\pi R}{2}$

SW0077

D. Standing waves

Paragraph for Question No. 36 to 38

In an organ pipe (may be closed or open) of length 1m standing wave is setup, whose equation for longitudinal displacement is given by $\xi = (0.1\text{ mm})\cos\frac{2\pi}{0.8}(y)\cos(400)\,t$ where y is measured from the top of the tube in meters and t in second.



36. The upper end and the lower ends of the tube are respectively:

(A) open – closed (B) closed – open

(C) open – open (D) closed – closed

SW0075

37. The air column is vibrating in

(A) First overtone (B) Second overtone

(C) Third harmonic (D) Fundamental mode

SW0075

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38. Equation of the standing wave in terms of excess pressure is (Bulk modulus of air B = 5×10^5 N/m^2)

(A) $P_{ex} = (125\ \pi N/m^2) \sin \frac{2\pi}{0.8} (y) \cos (400t)$

(B) $P_{ex} = (125\ \pi\ N/m^2) \cos \frac{2\pi}{0.8} (y) \sin (400t)$

(C) $P_{ex} = (225\ \pi N/m^2) \sin \frac{2\pi}{0.8} (y) \cos (200t)$

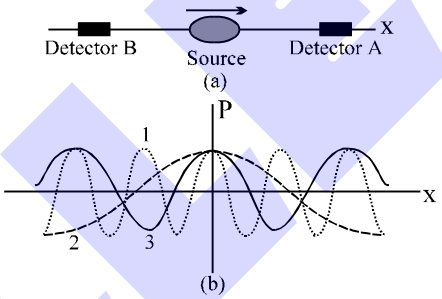
(D) $P_{ex} = (225\ \pi N/m^2) \cos \frac{2\pi}{0.8} (y) \sin (200t)$

SW0075

E. Doppler effect

Paragraph for Question No. 39 to 42 (4 questions)

A source emitting a sound wave at a certain frequency moves with constant speed along an x-axis figure (a). The source moves directly towards a stationary detector A and directly away from another stationary detector B. The superimposed three plots of figure (b) indicate the pressure function P(x) of the sound wave as measured by detector A, by detector B, and by someone (c) in the rest frame of the source.



39. Which of the following plot corresponds to the measurement done by detector A?

(A) 1 (B) 2

(C) 3 (D) These plots are not possible

SW0198

40. The plot corresponding to the measurement done by detector B is

(A) 1 (B) 2

(C) 3 (D) These plots are not possible

SW0199

41. The plot corresponding to the measurement done by the detector C is

(A) 1 (B) 2

(C) 3 (D) These plots are not possible

SW0200

42. Now the source stops and begins to move along y-axis with same speed, the plot which corresponds to the measurement of B now is

(A) 1 (B) 2 (C) 3 (D) none of these

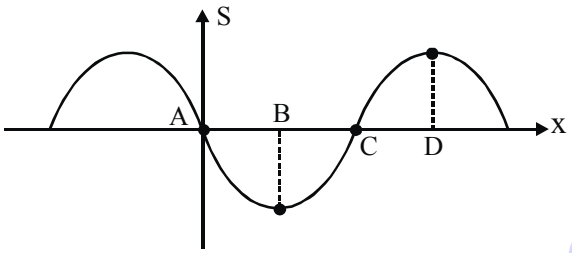
SW0201

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Waves-EM Wave & Wave Op\Eng\02_SW\01_TheEx.p65

MATRIX MATCH TYPE QUESTIONS

A. Displacement and pressure wave equation

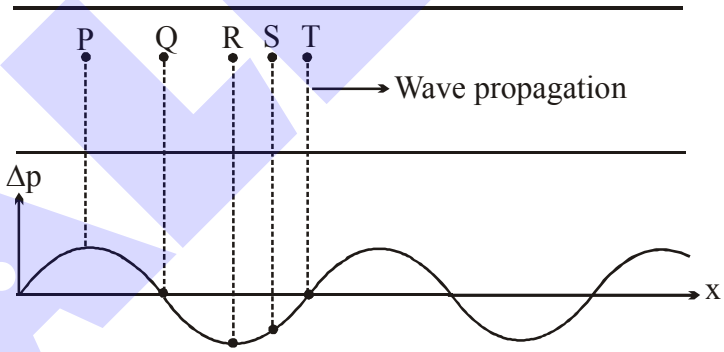
43. Figure shows a graph of particle displacement function of x at t = 0 for a longitudinal wave travelling in positive x-direction in a gas. A,B,C,D denote position of particles in space.



	Column-I		Column-II
(A)	point A	(P)	Particle velocity is in direction of wave propagation
(B)	point B	(Q)	Maximum magnitude of strain
(C)	point C	(R)	Excess pressure is zero
(D)	point D	(S)	Maximum density
		(T)	Maximum magnitude of excess pressure

SW0080

44. Sound is travelling in a long tube towards right and the graph of excess pressure variation versus position (at some instant) is given below.
Match velocities in column-I with column-II. P,Q,R,S,T are medium particles inside the tube.



	Column-I		Column-II
(A)	velocity is towards right	(P)	P
(B)	velocity is towards left	(Q)	Q
(C)	velocity is zero	(R)	R
(D)	Speed is maximum	(S)	S
		(T)	T

SW0081

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EXERCISE (S)

A. Displacement and pressure wave equation

1. The equation of a plane wave is given by $\Psi = (10^{-3}) \sin \pi (10t - \sqrt{3}\,x - y)$ where x, y and Ψ are expressed in metre and t in second. Find (a) the direction of propagation of the wave (b) wavelength and (c) phase difference between two points A ($\sqrt{3}$ m, 1m) and B($2\sqrt{3}$ m, 2m).

SW0202

B. Properties of sound

2. If the intensity is increased by a factor of 20, by how many decibels is the intensity level increased.

SW0203

3. The sound level at a point is increased by 40 dB. By what factor is the pressure amplitude increased?

SW0204

4. A 1000 m long rod of density 10.0×10^4 kg/m^3 and having young's modulus $Y = 10^{11}$ Pa, is clamped at one end. It is hammered at the other free end as shown in the figure. The longitudinal pulse goes to right end, gets reflected and again returns to the left end. How much time (in sec) the pulse take to go back to initial point?



SW0205

5. In a mixture of gases, the average number of degrees of freedom per molecule is 6. The rms speed of the molecules of the gas is c. Find the velocity of sound in the gas.

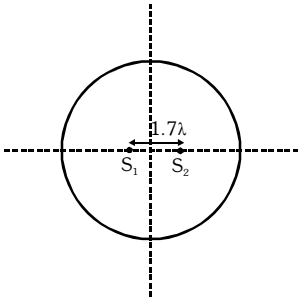
SW0002

6. The loudness level at a distance R from a long linear source of sound is found to be 40dB. At this point, the amplitude of oscillations of air molecules is 0.01 cm. Then find the loudness level & amplitude at a point located at a distance '10R' from the source.

SW0003

C. Interference of waves and beats

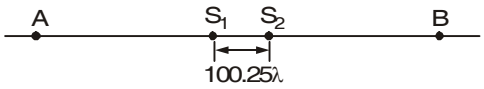
7. Find the number of maxima attend on circular perimeter as shown in the figure. Assume radius of circle >>>λ.



SW0206

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8. S_1 and S_2 are two coherent current sources of radiations separated by distance 100.25λ, where λ is the wave length of radiation. S_1 leads S_2 in phase by π/2. A and B are two points on the line joining S_1 and S_2 as shown in figure. The ratio of amplitudes of source S_1 and S_2 are in ratio 1 : 2. The ratio of intensity at A to that B $\left(\frac{I_B}{I_A}\right)$ is



SW0207

9. When two progressive waves $y_1 = 4\sin(2x - 6t)$ and $y_2 = 3\sin\left(2x - 6t - \frac{\pi}{2}\right)$ are superimposed, the amplitude of the resultant wave is **[IIT-JEE 2010]**

SW0087

10. A, B and C are three tuning forks. Frequency of A is 350Hz. Beats produced by A and B are 5 per second and by B and C are 4 per second. When a wax is put on A beat frequency between A and B is 2Hz and between A and C is 6Hz. Then, find the frequency of B and C respectively.

SW0012

11. A tuning fork having n = 158 Hz, produce 3 beats/s with another. As we file the arm of unknown, beats become 7 then calculate the frequency of unknown.

SW0208

12. Two identical piano wires have a fundamental frequency of 600 vib/sec, when kept under the same tension. What percentage increase in the tension of one wire will lead to the occurrence of six beats per second when both wires vibrate simultaneously.

SW0209

D. Standing waves

13. The equation of a longitudinal standing wave due to superposition of the progressive waves produced by two sources of sound is $s = -20 \sin 10\,\pi x \sin 100\,\pi t$ where s is the displacement from mean position measured in mm ; x is in meters and t in seconds. The specific gravity of the medium is 10^{-3}. Find

(a) wavelength, frequency and velocity of the progressive waves.

(b) Bulk modulus of the medium and the pressure amplitude of the progressive waves.

(c) minimum distance between pressure antinode and the displacement antinode.

SW0210

14. The air column in a pipe closed at one end and open to atmosphere at the other end is made to vibrate in its fifth harmonic by a tuning fork of frequency 470 Hz. The length of air column is $\frac{15}{16}$ m. Neglect end correction. Let p_0 denote the maximum gauge pressure at the closed end

(a) Find the speed of sound in air.

(b) Draw the graph of pressure amplitude vs distance from the open end of the tube.

(c) Find the points where the maximum gauge pressure is $\frac{p_0}{2}$.

SW0211

node06\B09C-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_Th+Ex_p65

15. The first overtone frequency of a closed organ pipe P_1 is equal to the fundamental frequency of an open organ pipe P_2. If the length of the pipe P_1 is 30 cm, what will be the length of P_2?

SW0212

16. The first overtone of a pipe closed at one end resonates with the third harmonic of a string fixed at its ends. The ratio of the speed of sound to the speed of transverse wave travelling on the string is 2 :1. Find the ratio of the length of pipe to the length of string.

SW0004

17. A tuning fork of frequency 480 Hz resonates with a tube closed at one end of length, 16 cm and diameter 5 cm in fundamental mode. Calculate velocity of sound in air. **[JEE 2003]**

SW0006

18. In a resonance-column experiment, a long tube, open at the top, is clamped vertically. By a separate device, water level inside the tube can be moved up or down. The section of the tube from the open end to the water level act as a closed organ pipe. A vibrating tuning fork is held above the open end, first and the second resonances occur when the water level is 24.1 cm and 74.1 cm respectively below the open end. Find the diameter of the tube.

SW0005

19. A 3.6 m long vertical pipe is filled completely with a liquid. A small hole is drilled at the base of the pipe due to which liquids starts leaking out. This pipe resonates with a tuning fork. The first two resonances occur when height of water column is 3.22 m and 2.34 m respectively. The area of cross-section of pipe is $(20\pi)A_0 cm^2$. Find the vlaue of A_0.

SW0213

20. AB is a cylinder of length 1 m fitted with a thin flexible diaphram C at middle and two other thin flexible diaphram A and B at the ends. The portions AC and BC contain hydrogen and oxygen gases respectively. The diaphragms A and B are set into vibrations of the same frequency. What is the minimum frequency of these vibrations for which diaphram C is a node ? Under the condition of the experiment the velocity of sound in hydrogen is 1100 m/s and oxygen 300 m/s.

SW0214

21. A steel rod having a length of 1 m is fastened at its middle. Assuming young's modulus to be 2×10^{11} Pa, and density to be 8 gm/cm^3 find the fundamental frequency of the longitudinal vibration and frequency of first overtone.

SW0008

E. Doppler effect

22. Two trains travelling in opposite directions at 126 km/hr each, cross each other while one of them is whistling. If the frequency of the node is 2.22 kHz find the apparent frequency as heard by an observer in the other train :

(a) Before the trains cross each other

(b) After the trains have crossed each other. (v_{sound} = 335m/sec)

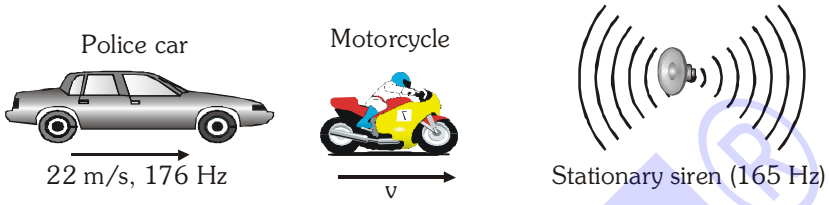
SW0215

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_TheEx.p65

23. A bat is flitting about in a cave, navigating via ultrasonic beeps. Assume that the sound emission frequency of the bat is 40 kHz. During one fast swoop directly toward a flat wall surface, the bat is moving at 0.03 times the speed of sound in air. What frequency does the bat hear reflected off the wall?

SW0216

24. A police car moving at 22 m/s chases a motocyclist. The police man sounds his horn at 176 Hz, while both of them move towards a stationary siren of frequency 165 Hz. Calculate the speed of the motorcycle. If it is given that the motorcyclist does not observe any beats :–

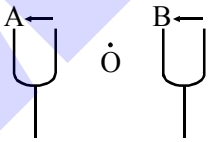


SW0217

25. A car is moving towards a huge wall with a speed = c/10, where c = speed of sound in still air. A wind is also blowing parallel to the velocity of the car in the same direction and with the same speed. If the car sounds a horn of frequency f, then what is the frequency of the reflected sound of the horn heard by driver of the car?

SW0016

26. Two tuning forks A and B each of natural frequency 85 Hz move with velocity 10 m/s relative to stationary observer 'O'. Fork A moves away from the observer while the fork B moves towards him as shown in the figure. A wind with a speed 10 m/s is blowing in the direction of motion of fork A. Find the beat frequency measured by the observer (in Hz). [Take speed of sound in air as 340 m/s]



SW0218

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EXERCISE - JM

1. A pipe of length 85 cm is closed from one end. Find the number of possible natural oscillations of air column in the pipe whose frequencies lie below 1250 Hz. The velocity of sound in air is 340 m/s.

[JEE Main - 2014]

(1) 6 (2) 4 (3) 12 (4) 8

SW0083

2. A train is moving on a straight track with speed 20 ms^{-1}. It is blowing its whistle at the frequency of 1000 Hz. The percentage change in the frequency heard by a person standing near the track as the train passes him is (speed of sound = 320 ms^{-1}) close to :- **[JEE Main - 2015]**

(1) 18% (2) 24% (3) 6% (4) 12%

SW0084

3. A pipe open at both ends has a fundamental frequency f in air. The pipe is dipped vertically in water so that half of it is in water. The fundamental frequencty of the air column is now :-

[JEE-Main-2016]

(1) f (2) $\frac{f}{2}$ (3) $\frac{3f}{4}$ (4) 2f

SW0085

4. A granite rod of 60 cm length is clamped at its middle point and is set into longitudinal vibrations. The density of granite is 2.7×10^3 kg/m^3 and its Young's modulus is 9.27×10^{10} Pa. What will be the fundamental frequency of the longitudinal vibrations ? **[JEE-Main-2018]**

(1) 2.5 kHz (2) 10 kHz (3) 7.5 kHz (4) 5 kHz

SW0113

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

5. Two coherent sources produce waves of different intensities which interfere. After interference, the ratio of the maximum intensity to the minimum intensity is 16. The intensity of the waves are in the ratio: **[JEE-Main-2019_Jan]**

(1) 4 : 1 (2) 25 : 9 (3) 16 : 9 (4) 5 : 3

SW0114

6. A resonance tube is old and has jagged end. It is still used in the laboratory to determine velocity of sound in air. A tuning fork of frequency 512 Hz produces first resonance when the tube is filled with water to a mark 11 cm below a reference mark, near the open end of the tube. The experiment is repeated with another fork of frequency 256 Hz which produces first resonance when water reaches a mark 27 cm below the reference mark. The velocity of sound in air, obtained in the experiment, is close to: **[JEE-Main-2019_Jan]**

(1) 328ms^{-1} (2) 322ms^{-1} (3) 341ms^{-1} (4) 335ms^{-1}

SW0115

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7. A person standing on an open ground hears the sound of a jet aeroplane, coming from north at an angle 60° with ground level. But he finds the aeroplane right vertically above his position. If υ is the speed of sound, speed of the plane is : **[JEE-Main-2019_Jan]**

(1) $\frac{2\upsilon}{\sqrt{3}}$ (2) υ (3) $\frac{\upsilon}{2}$ (4) $\frac{\sqrt{3}}{2}\upsilon$

SW0116

8. The pressure wave, $P = 0.01 \sin [1000t - 3x]\ Nm^{-2}$, corresponds to the sound produced by a vibrating blade on a day when atmospheric temperature is 0°C. On some other day, when temperature is T, the speed of sound produced by the same blade and at the same frequency is found to be 336 ms^{-1} . Approximate value of T is : **[JEE-Main-2019_April]**

(1) 15°C (2) 12°C (3) 4°C (4) 11°C

SW0117

9. A tuning fork of frequency 480 Hz is used in an experiment for measuring speed of sound (ν) in air by resonance tube method. Resonance is observed to occur at two successive lengths of the air column, $l_1 = 30$ cm and $l_2 = 70$ cm. Then ν is equal to : **[JEE-Main-2019_April]**

(1) 332 ms^{-1} (2) 379 ms^{-1} (3) 384 ms^{-1} (4) 338 ms^{-1}

SW0118

10. A small speaker delivers 2 W of audio output. At what distance from the speaker will one detect 120dB intensity sound ? [Given reference intensity of sound as $10^{-12} W/m^2$] **[JEE-Main-2019_April]**

(1) 10 cm (2) 30 cm (3) 40 cm (4) 20 cm

SW0119

11. A submarine (A) travelling at 18 km/hr is being chased along the line of its velocity by another submarine (B) travelling at 27 km/hr. B sends a sonar signal of 500 Hz to detect A and receives a reflected sound of frequency ν. The value of ν is close to : (Speed of sound in water = 1500 ms^{-1})

(1) 499 Hz (2) 502 Hz (3) 507 Hz (4) 504 Hz

SW0120

12. A one metre long (both ends open) organ pipe is kept in a gas that has double the density of air at STP. Assuming the speed of sound in air at STP is 300 m/s, the frequency difference between the fundamental and second harmonic of this pipe is _____ Hz. **[JEE-Main-2020_Jan]**

SW0121

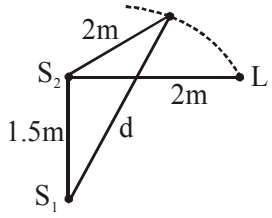
13. Assume that the displacement(s) of air is proportional to the pressure difference (Δp) created by a sound wave. Displacement(s) further depends on the speed of sound (v), density of air (ρ) and the frequency (f). If Δp~10Pa, v~300 m/s, p~1 kg/m^3 and f~1000Hz, then s will be the order of (take multiplicative constant to be 1) **[JEE-Main-2020_Sep]**

(1) 10 mm (2) $\frac{3}{100}$ mm (3) 1 mm (4) $\frac{1}{10}$ mm

SW0124

node06\B09C-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_Th+Ex.p65

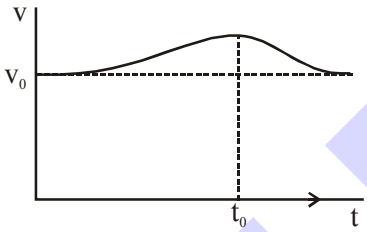
14. Two coherent sources of sound, S_1 and S_2, produce sound waves of the same wavelength, $\lambda = 1$ m, in phase. S_1 and S_2 are placed 1.5 m apart (see fig.) A listener, located at L, directly in front of S_2 finds that the intensity is at a minimum when he is 2m away from S_2. The listener moves away from S_1, keeping his distance from S_2 fixed. The adjacent maximum of intensity is observed when the listener is at a distance d from S_1. Then, d is : **[JEE-Main-2020_Sep]**

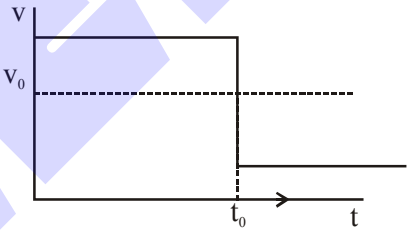


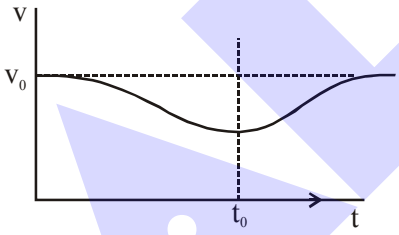
(1) 12m (2) 3m (3) 5m (4) 2m

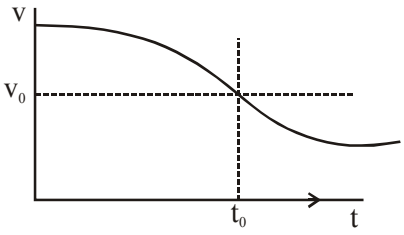
SW0125

15. A sound source S is moving along a straight track with speed v, and is emitting sound of frequency v_0 (see figure). An observer is standing at a finite distance, at the point O, from the track. The time variation of frequency heard by the observer is best represented by :
(t_0 represents the instant when the distance between the source and observer is minimum)
[JEE-Main-2020_Sep]

(1)


(2)


(3)


(4)


SW0126

16. A wire of density 9×10^{-3} kg cm^{-3} is stretched between two clamps 1 m apart. The resulting strain in the wire is 4.9×10^{-4}. The lowest frequency of the transverse vibrations in the wire is (Young's modulus of wire $Y = 9 \times 10^{10}$ Nm^{-2}), (to the nearest integer),_____. **[JEE-Main-2020_Sep]**

SW0127

17. A student is performing the experiment of resonance column. The diameter of the column tube is 6cm. The frequency of the tuning fork is 504 Hz. Speed of the sound at the given temperature is 336 m/s. The zero of the meter scale coincides with the top end of the resonance column tube. The reading of the water level in the column when the first resonance occurs is: **[JEE-Main-2021_Feb]**

(1) 13 cm (2) 16.6 cm (3) 18.4 cm (4) 14.8 cm

SW0219

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_TheEx.p65

18. A tuning fork A of unknown frequency produces 5 beats/s with a fork of known frequency 340 Hz. When fork A is filed, the beat frequency decreases to 2 beats/s. What is the frequency of fork A ?

[JEE-Main-2021_Feb]

(1) 342 Hz (2) 345 Hz (3) 335 Hz (4) 338 Hz

SW0220

19. A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with same frequency. The length of the open pipe is $\frac{x}{3}L\sqrt{\frac{\rho_1}{\rho_2}}$ where x is _____. (Round off to the Nearest Integer)

[JEE-Main-2021_March]

SW0221

20. With what speed should a galaxy move outward with respect to earth so that the sodium-D line at wavelength 5890 Å is observed at 5896 Å ?

[JEE-Main-2021_July]

(1) 306 km/sec (2) 322 km/sec (3) 296 km/sec (4) 336 km/sec

SW0222

21. A tuning fork is vibrating at 250 Hz. The length of the shortest closed organ pipe that will resonate with the tuning fork will be _____ cm. (Take speed of sound in air as 340 ms^{-1})

[JEE-Main-2021_Aug]

SW0223

22. The velocity of sound in a gas. in which two wavelengths 4.08m and 4.16m produce 40 beats in 12s, will be :

[JEE-Main-2022_June]

(A)2.82.8 ms^{-1} (B) 175.5 ms^{-1} (C) 353.6 ms^{-1} (D) 707.2 ms^{-1}

SW0224

23. In an experiment to determine the velocity of sound in air at room temperature using a resonance is observed when the air column has a length of 20.0 cm for a tuning fork of frequency 400 Hz is used. The velocity of the sound at room temperature is 336 ms^{-1}. The third resonance is observed when the air column has a length of ____cm.

[JEE-Main-2022_June]

SW0225

24. Sound travels in a mixture of two moles of helium and n moles of hydrogen. If rms speed of gas molecules in the mixture is $\sqrt{2}$ times the speed of sound, then the value of n will be

[JEE-Main-2022_July]

(A) 1 (B) 2 (C) 3 (D) 4

SW0226

25. The frequency of echo will be _____ Hz if the train blowing a whistle of frequency 320 Hz is moving with a velocity of 36 km/h towards a hill from which an echo is heard by the train driver. Velocity of sound in air is 330 m/s.

[JEE-Main-2022_July]

SW0227

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EXERCISE (JA)

1. A student is performing an experiment using a resonance column and a tuning fork of frequency 244 s^{-1}. He is told that the air in the tube has been replaced by another gas (assume that the column remains filled with the gas). If the minimum height at which resonance occurs is (0.350 ± 0.005) m, the gas in the tube is

(**Useful information :** $\sqrt{167RT} = 640\ J^{1/2}\ mole^{-1/2}$; $\sqrt{140RT} = 590\ J^{1/2}\ mole^{-1/2}$. The molar masses M in grams are given in the options. Take the values of $\sqrt{\frac{10}{M}}$ for each gas as given there.)

[JEE Advanced-2014]

(A) Neon $\left(M = 20, \sqrt{\frac{10}{20}} = \frac{7}{10}\right)$ (B) Nitrogen $\left(M = 28, \sqrt{\frac{10}{28}} = \frac{3}{5}\right)$

(C) Oxygen $\left(M = 32, \sqrt{\frac{10}{32}} = \frac{9}{16}\right)$ (D) Argon $\left(M = 36, \sqrt{\frac{10}{36}} = \frac{17}{32}\right)$

SW0092

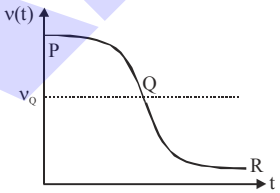
2. Four harmonic waves of equal frequencies and equal intensities I_0 have phase angles 0, $\pi/3$, $2\pi/3$ and π. When they are superposed, the intensity of the resulting wave is nI_0. The value of n is.

[JEE-Advance-2015]

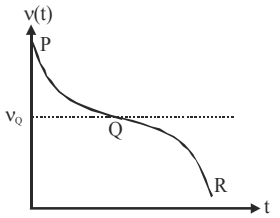
SW0093

3. Two loudspeakers M and N are located 20m apart and emit sound at frequencies 118 Hz and 121 Hz, respectively. A car is initially at a point P, 1800 m away from the midpoint Q of the line MN and moves towards Q constantly at 60 km/hr along the perpendicular bisector of MN. It crosses Q and eventually reaches a point R, 1800 m away from Q. Let $\nu(t)$ represent the beat frequency measured by a person sitting in the car at time t. Let ν_P, ν_Q and ν_R be the beat frequencies measured at locations P, Q and R, respectively. The speed of sound in air is 330 ms^{-1}. Which of the following statement(s) is(are) true regarding the sound heard by the person ? **[JEE Advanced 2016]**

(A) The plot below represents schematically the variation of beat frequency with time



(B) The plot below represents schematically the variations of beat frequency with time



(C) The rate of change in beat frequency is maximum when the car passes through Q

(D) $\nu_P + \nu_R = 2\nu_Q$

SW0094

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\4-Wave on String-Sound Wave-EM Wave & Wave Op\Eng\02_SW\01_TheEx.p65

4. A stationary source emits sound of frequency $f_0 = 492$ Hz. The sound is reflected by a large car approaching the source with a speed of 2 ms^{-1}. The reflected signal is received by the source and superposed with the original. What will be the beat frequency of the resulting signal in Hz ? (Given that the speed of sound in air is 330 ms^{-1} and the car reflects the sound at the frequency it has received).

[JEE Advanced 2017]

SW0095

5. Two men are walking along a horizontal straight line in the same direction. The man in front walks at a speed 1.0 ms^{-1} and the man behind walks at a speed 2.0 ms^{-1}. A third man is standing at a height 12m above the same horizontal line such that all three men are in a vertical plane. The two walking men are blowing identical whistles which emit a sound of frequency 1430 Hz. The speed of sound in air is 330 ms^{-1}. At the instant, when the moving men are 10 m apart, the stationary man is equidistant from them. The frequency of beats in Hz, heard by the stationary man at this instant, is _____

[JEE Advanced 2018]

SW0096

6. In an experiment to measure the speed of sound by a resonating air column, a tuning fork of frequency 500 Hz is used. The length of the air column is varied by changing the level of water in the resonance tube. Two successive resonances are heard at air columns of length 50.7 cm and 83.9 cm. Which of the following statements is (are) true ? **[JEE Advanced 2018]**

(A) The speed of sound determined from this experiment is 332 ms^{-1}

(B) The end correction in this experiment is 0.9 cm

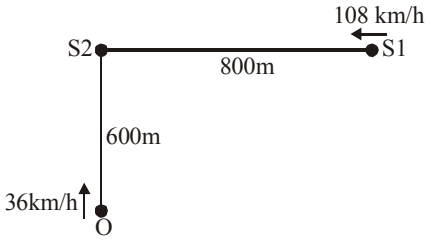
(C) The wavelength of the sound wave is 66.4 cm

(D) The resonance at 50.7 cm corresponds to the fundamental harmonic

SW0097

7. A train S1, moving with a uniform velocity of 108 km/h, approaches another train S2 standing on a platform. An observer O moves with a uniform velocity of 36 km/h towards S2, as shown in figure. Both the trains are blowing whistles of same frequency 120 Hz. When O is 600 m away from S2 and distance between S1 and S2 is 800 m, the number of beats heard by O is ____.

[Speed of the sound = 330 m/s] **[JEE Advanced 2019]**



SW0098

8. A stationary tuning fork is in resonance with an air column in a pipe. If the tuning fork is moved with a speed of 2 ms^{-1} in front of the open end of the pipe and parallel to it, the length of the pipe should be changed for the resonance to occur with the moving tuning fork. If the speed of sound in air is 320 ms^{-1}, the smallest value of the percentage change required in the length of the pipe is _____.-

[JEE Advanced 2020]

SW0128

9. A source, approaching with speed u towards the open end of a stationary pipe of length L, is emitting a sound of frequency f_s. The farther end of the pipe is closed. The speed of sound in air is v and f_0 is the fundamental frequency of the pipe. For which of the following combination(s) of u and fs, will the sound reaching the pipe lead to a resonance ? **[JEE Advanced 2021]**

(A) u = 0.8 v and $f_s = f_0$

(B) u = 0.8 v and $f_s = 2f_0$

(C) u = 0.8 v and $f_s = 0.5\ f_0$

(D) u = 0.5 v and $f_s = 1.5\ f_0$

SW0129

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	B	A	B	B	A	A	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	C	A	A	B	A	B	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	D	A	A	B	A	A	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	A	A	A	A	B	A	D	A	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	B	C	A	C	D	C	B	A	A
Que.	51	52	53	54	55	56	57	58		
Ans.	B	B	A	A	A	B	A	D		

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	D	C	B	C	B	B	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	A	A	B	C	C	A	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A,C,D	A,B,C,D	A,C	B,C	A,B,D	B,D	B,C	B,D	A,B,D	B,D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A,B,C	A,C,D	B	A	B	A	B	A	A	B
Que.	41	42	43				44			
Ans.	C	D	(A)-P,Q,S,T; (B)-R; (C)-Q,T; (D)-R				(A)-P; (B)-R,S; (C)-Q,T; (D)-P,R			

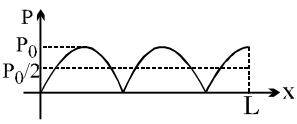
EXERCISE # S

1. (a) Along a straight in xy plane through origin in at 30° with x-axis, (b) 1m, (c) 4π

2. 13 dB **3.** ≈ 100 **4.** 2 **5.** 2c/3 **6.** 30 dB, $10\sqrt{10}$ μm

7. 6 **8.** 9 **9.** 5 **10.** 345, 341 or 349 Hz **11.** 161 Hz **12.** 2%

13. (a) 1/5 m, 50 Hz, 10 m/s; (b) 100 Pa, 10π Pa, (c) 1/20 m

14. (a) 352.5 m/s, (b) 

, (c) $\frac{l}{15}, \frac{5l}{15}, \frac{7l}{15}, \frac{11l}{15}, \frac{13l}{15}$ **15.** 20 cm

16. 1 : 1 **17.** 336 m/s **18.** 3 cm **19.** 5 **20.** 1650 Hz

21. 2.5 kHz, 7.5 kHz **22.** (a) 2738 Hz; (b) 1800 Hz **23.** 42.47 kHz

24. 22 m/s **25.** 11f / 9 **26.** 5

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	1	4	2	1	3	3	3	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	106.00 to 107.20	2	2	4	35.00	4	3	4	1
Que.	21	22	23	24	25					
Ans.	34	D	104	B	340					

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9
Ans.	D	3	A,C,D	6	5 [4.99, 5.01]	A,B,C	8.12 to 8.13	0.62 to 0.64	A,D

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